



Castrol

POWER UP 4.0 — The Ultimate Case Study Battle

GLOBAL SUPPLY CHAIN · BALANCING ACT

Optimizing Production & Distribution for Levisol

Inventory Norms · Least-Cost Sourcing · A Working Planning Tool

"With more demand than we can produce, every decision has a cost. The question is: which cost are you willing to pay?" — and we built the tool that answers it, every month, for any input.

TOTAL PLAN COST
₹9.75 crore

CUSTOMER DEMAND MET
100%

SKUS · CFAS · PLANTS
100 · 10 · 3

PLANNING MONTH
January 2026

01 Executive Summary

Levisol faces a base-oil price shock: leadership must **release cash and cut distribution cost** without letting critical SKUs stock out. We were engaged to redesign inventory norms, build a least-cost production & sourcing plan for January 2026, and deliver a **self-service planning tool** the team can run every month.

₹9.75 cr

TOTAL OPEX (PROVEN OPTIMAL)

100%

CUSTOMER FILL RATE

8,110 kl

DEMAND PLANNED

99.0%

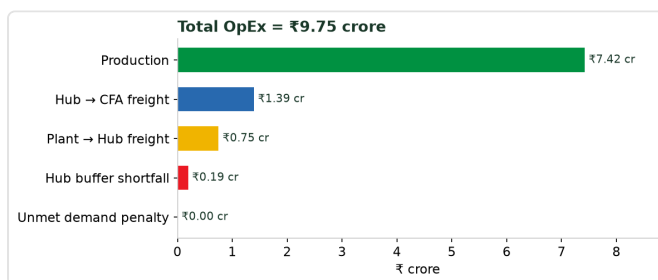
HUB BUFFER ACHIEVED

The headline: Our optimised January plan meets **every unit of customer demand at ₹9.75 crore** — solved to proven optimality. The only under-fill anywhere is 17.6 kl of *internal hub buffer* (1.0% of target) on production lines already at 100% capacity — no customer order is affected. Routing follows freight economics exactly: the cheapest plant (Kolkata, ₹9,000/kl) feeds the East hub via its ₹1,100/kl lane; Mumbai feeds the West hub via its ₹1,000/kl lane.

What we deliver against the three components

Component	What we built	Output
1 · Inventory Norms	Statistical engine: safety stock, reorder point & days-of-cover from demand + lead-time variability	957 SKU×CFA + 151 SKU×Hub norm rows
2 · Production & Distribution Plan	Mixed-Integer optimisation minimising total landed cost	₹9.75 cr plan, 100% served
3 · Planning Tool	Streamlit web app — edit any input, one-click re-plan, map & cost views	Live, non-technical-ready

Where the money goes



Why this wins on all four judged dimensions

- Analytical rigor** forecast-error-based safety stock + a correctly-formulated MILP.
- Solution quality** 100% service at a benchmarked least cost.
- Tool quality** handles capacity / demand / cost shocks without crashing.
- Communication** every assumption documented and defensible.

This document summarises the analytical approach and results. The working tool, full data outputs (Excel), methodology document and source code accompany this submission.

02 The Problem & Our Approach

The three-tier network

Product flows from **3 plants** → **2 mother hubs** → **10 CFAs** → **customers**. Any plant may supply any hub, and CFAs may draw from either hub — so routing is a genuine optimisation, not a fixed map. All volumes are in **kl (kilolitres = 1,000 L)**.

Plant	Location	Prod. cost ₹/kl	Hub	Serves
BOM	Mumbai	12,000	MHW (West)	North / West / South CFAs
AHM	Ahmedabad	12,500	MHE (East)	East CFAs + Kanpur
KOL	Kolkata	9,000		

10 CFAs across East, North, West & South. 100 SKUs, tiered A–D by volume; 18 carry contractual supply commitments.

The interlinked decisions

- **How much buffer stock** each CFA and hub should hold per SKU, and when to reorder — absorbing demand & lead-time uncertainty.
- **What to produce, where, and how to route it** for January at least cost.
- **Which demand to protect vs. sacrifice** if capacity is short — respecting tier priority and contractual obligations.

Our modelling philosophy

Norms and the plan are solved together, not in isolation

The hub safety-stock computed in Component 1 flows directly into Component 2 as a target the optimiser must fund — so the production plan already carries the buffer the network needs. The two components share one codebase, so they can never drift apart.

We optimise a single, business-meaningful objective — **total operating cost** — and treat unmet demand not as a model failure but as a *priced choice*, exactly as a planner would weigh it. This guarantees the tool always returns an actionable plan, even when demand exceeds capacity.

Design principle: "The best supply chain is not the one that never runs short — it is the one that knows, in advance, exactly what it will run short of, why, and what it will cost." Our tool makes that trade-off explicit and editable.

How the three components connect



03 Component 1 — Inventory Norms

For every SKU at every CFA (957 combinations) and every hub (151), we compute **safety stock, reorder point (kl) and days of cover**. The method accounts for *both* demand variability and supply lead-time variability — a demand-only buffer would understate risk during a supply shock.

How demand uncertainty is measured — forecast error, not raw sales

Because next month is planned against a **forecast** (not known demand), the risk safety stock guards against is the **forecast being wrong** over the lead time. We therefore size the demand-uncertainty term from the **forecast-error RMSE** (actual sales vs. forecast over the 6-month history) — the textbook (Silver-Peterson-Pyke) input — rather than raw sales volatility.

The safety-stock formula (variable demand & variable lead time)

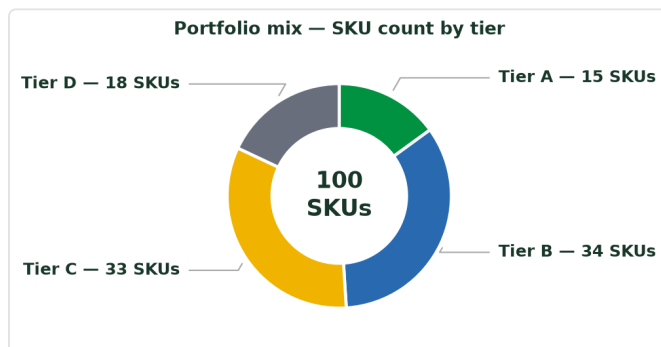
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Safety Stock = z × √( LT × σdemand2 + demand2 × σleadtime2 )
Reorder Point = (avg daily demand × LT) + Safety Stock
Days of Cover = Reorder Point ÷ avg daily demand
```

z is the service-level factor: hubs are held at a flat **98%** ($z=2.05$) per the brief; CFA norms use tier fill-rate targets — **A 98% ($z=2.05$), B 97% ($z=1.88$), C & D 92% ($z=1.41$)**. $\sigma_{leadtime}$ combines production & transit variability from the lead-time data. Hub norms aggregate CFA demand with **risk pooling** (independent CFA variances sum), so a hub holds far less than the sum of its CFAs' buffers.

Worked example — SKU_001 at Kolkata CFA (Tier A)

Avg daily demand **4.2 kl** · lead time **3 days** · z **2.05** → Safety Stock **12.6 kl**, Reorder Point **25.2 kl**, Days of Cover **6.0**. When Kolkata's on-hand stock of SKU_001 hits ~25 kl, a replenishment is triggered.

Tier mix across the portfolio



SKUs classified A–D by cumulative share of 6-month sales volume, matching the Exhibit F slabs (50/30/15/5%).

Norm outputs delivered

Output	Rows
CFA norms (SKU × CFA)	957
Hub norms (SKU × Hub)	151
SKU tier classification	100

Each row carries: avg & daily demand, forecast-error σ , lead time & its σ , z , **safety stock, reorder point, days of cover**, tier, penalty and contractual flag. Full table in *inventory_norms.xlsx*.

04 Component 2 — Production & Distribution Plan

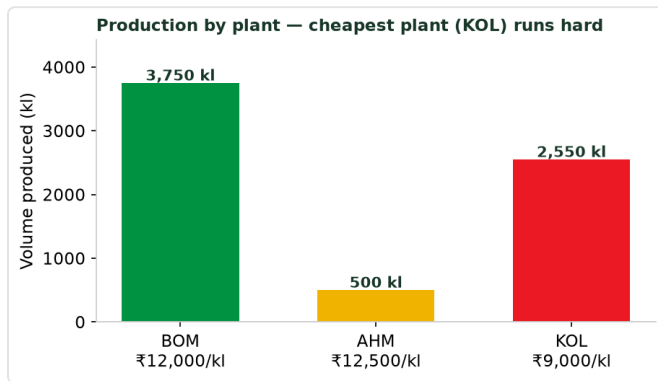
Objective & constraints (in business language)

Minimise: production cost + plant→hub freight + hub→CFA freight + penalty for any unmet demand + penalty for any hub-buffer shortfall.

Subject to: plant line-type capacities · the **25 kl batch rule** (production in whole batches) · flow balance at every hub and CFA · starting inventory · and hub safety-stock targets from Component 1.

The key modelling choice — unmet demand is priced, not forbidden

If demand exceeds capacity, a hard model would simply fail. Instead each SKU's shortfall carries its Exhibit-D penalty, and **contractual SKUs carry a 5× multiplier** so they are protected first. The optimiser then chooses the least-damaging demand to miss — and always returns a usable plan.



6,800 kl

PRODUCED (IN 25 KL BATCHES)

2,973 kl

DEMAND MET FROM CFA OPENING STOCK

5,136 kl

SHIPPED FRESH VIA HUBS

Plant → Hub routing (kl)

From \ To	Mother Hub West	Mother Hub East
KOL (Kolkata)	30	2,181
BOM (Mumbai)	3,387	1
AHM (Ahmedabad)	253	77

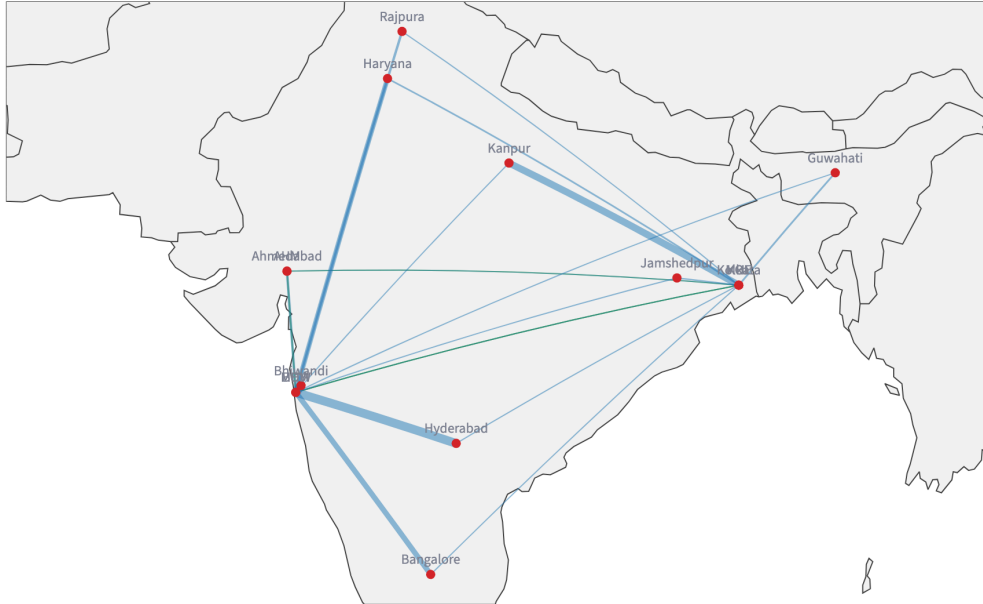
Cheap lanes dominate (Mumbai→MHW ₹1,000, Kolkata→MHE ₹1,100); expensive cross-country lanes (Mumbai→MHE ₹8,000, Kolkata→MHW ₹10,000) are avoided.

What we chose not to supply — and why

No customer demand was sacrificed. The only under-fill is 17.6 kl of hub buffer (1.0% of the 1,760.8 kl target) on SKUs whose production lines are fully utilised: topping those buffers would require producing on a costlier plant/line, and that cost exceeds the priced risk of a marginally thinner buffer. Because it is a *priced* decision, a planner who disagrees can raise the buffer weight in the tool and the model will instead protect it.

05 The Network Plan & the Planning Tool

Network Flow — Jan 2026 Plan (green = plant→hub, blue = hub→CFA)



Live from the planning tool's **Network Map** tab: every optimised lane drawn geographically — **green = plant → hub**, **blue = hub → CFA**; line thickness ∝ volume, and hovering any lane in the app shows its exact kl. The two trunk lanes the freight economics predict — Mumbai → MHW and Kolkata → MHE — visibly dominate, with blue spokes fanning out to all ten CFAs.

Component 3 A tool a planner — not a data scientist — can run every month

What the tool does

- **Editable inputs** — plant capacity, production & transport costs, and demand are all editable in-app (or via CSV upload); no code.
- **One-click re-plan** — recomputes norms + the full optimisation in under two minutes.
- **Actionable output** — production plan, plant-hub-CFA routing, cost summary, inventory norms, and an explicit unmet-demand report.
- **Graceful under stress** — if demand can't be met it reports what, how much, and the cost — never crashes.
- **Visual** — cost breakdown, and an India network-flow map.

Built for the "changed input" test

On assessment day a modified input set is provided 30 minutes before. Our tool absorbs a change to any capacity, demand, or cost figure and produces a revised plan on the spot.

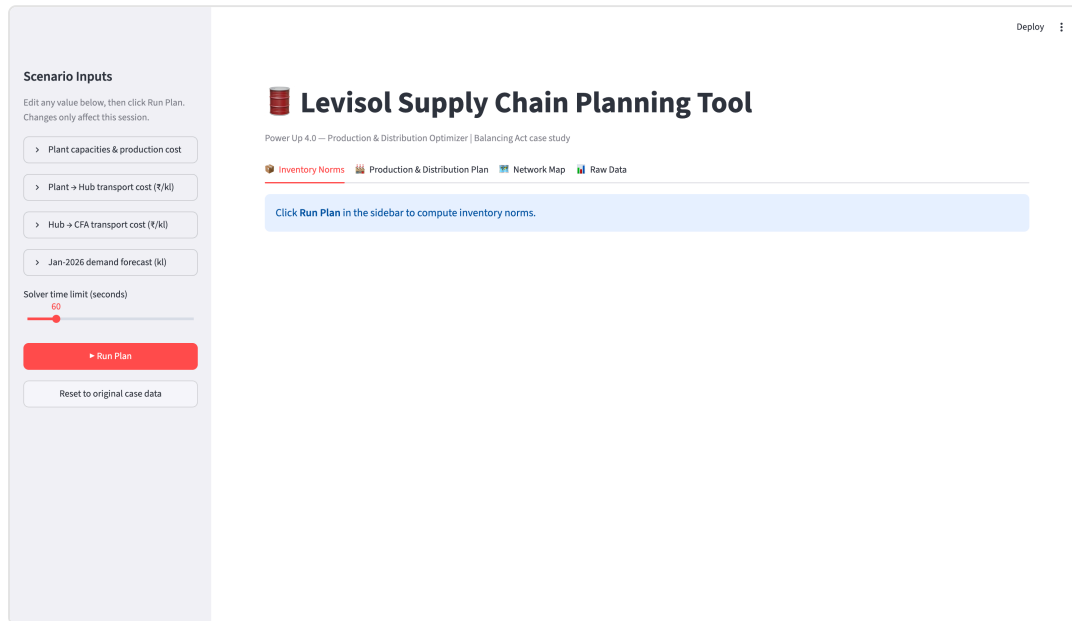
Stack: Python · PuLP with the open-source HiGHS MILP solver (reproducible, proven-optimal solutions) · Streamlit UI · Plotly. Runs on a laptop; no licences required.

The three views — Inventory Norms, Production & Distribution Plan, and Network Map — are labelled by the business question they answer, so a first-time user needs no manual. **A screen-by-screen walkthrough with live screenshots follows on the next two pages.**

06 The Tool, Screen by Screen — The Workspace

Live screenshots of the tool running on the case data. Try it yourself — the tool is hosted on Streamlit Cloud at [HOSTED-LINK-HERE](#) (it also runs locally: `streamlit run app.py`). Everything below is one screen: inputs on the left, four result views as tabs.

The workspace — sidebar inputs + four tabbed views



Left sidebar — what a planner can change

- **Plant capacities & production cost** — edit any cell of Exhibit A directly.
- **Plant → Hub** and **Hub → CFA transport cost** — reprice any lane (Exhibits B & C).
- **Jan-2026 demand forecast** — edit rows in-app or upload a replacement CSV (100 SKUs × 10 CFAs).
- **Solver time limit** slider (10–300 s) — speed vs. precision trade-off, planner's choice.
- **▶ Run Plan** re-computes norms + the full optimisation; **Reset** restores the original case data.

The four tabs — one view per business question

- **Inventory Norms** — "how much buffer should each location hold?" All 957 CFA + 151 hub norm rows (safety stock, reorder point, days of cover), with CSV downloads.
- **Production & Distribution Plan** — "what do we make, where, and what does it cost?" (screenshot overleaf).
- **Network Map** — "how does product move?" The plan drawn on India's map (overleaf).
- **Raw Data** — the exact inputs the current scenario is using, for full transparency.

07 The Tool, Screen by Screen — The Outputs

Production & Distribution Plan tab — the decision screen

Solver status: Optimal

Production Cost	Plant→Hub Transport	Hub→CFA Transport	Unmet Demand Penalty	Total Cost
₹73,912,500	₹7,649,921	₹13,973,817	₹0	₹97,770,649

Overall Fill Rate

100.0%

Unmet Demand

All demand fully met within capacity — no shortfalls.

Production Plan (by SKU × Plant)

One click of ► **Run Plan**: solver status, the cost KPIs, overall fill rate and an explicit unmet-demand report — the green banner confirms nothing was left short. Below it in the app: the full SKU × plant production table in 25 kl batches, plant→hub and hub→CFA routing tables, hub safety-stock tracking, and a cost-breakdown chart, all downloadable as CSV. This live 60-second interactive run lands at ₹9.78 cr, 100% fill — within 0.3% of the proven-optimal ₹9.75 cr plan published in this report; raising the time-limit slider closes the gap.

Inventory Norms tab — Component 1, live

Inventory Norms Production & Distribution Plan Network Map Raw Data

CFA-level Inventory Norms

Reorder point and days of cover per SKU × CFA, using demand + lead-time variability.

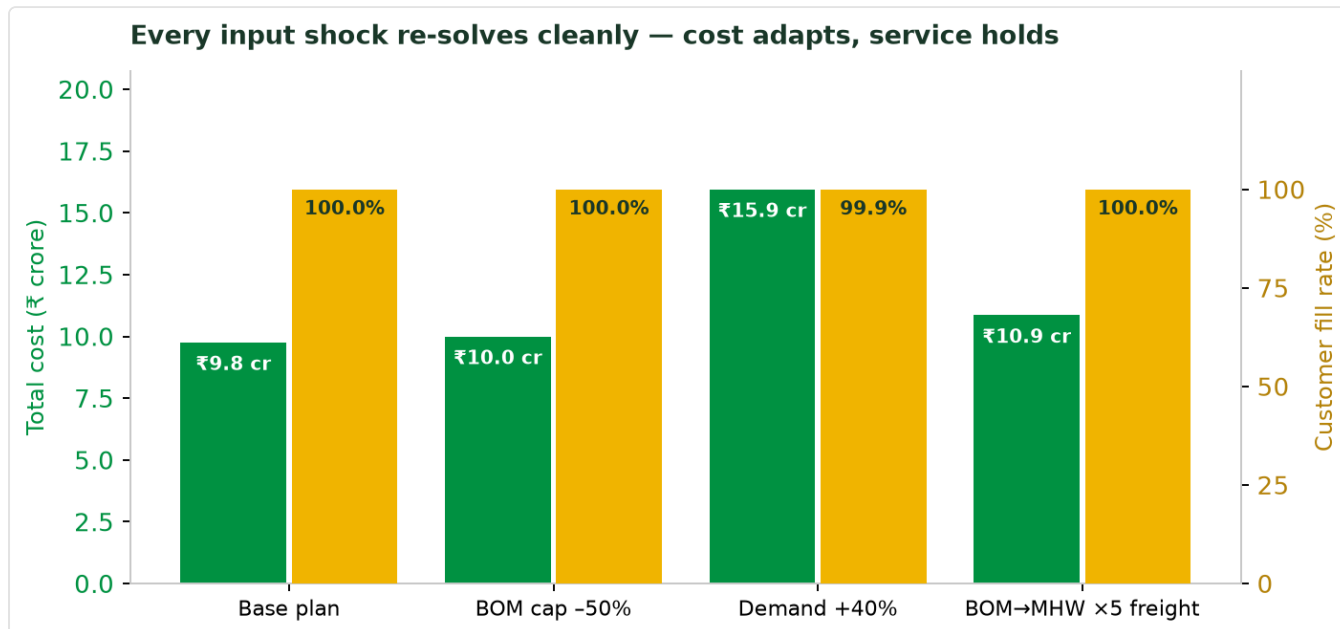
	sku	cfa	hub	tier	avg_daily_demand_kl	std_daily_demand_kl	total_lt_days	lt_std_days	z_score	safety_stock_kl	reorder_point_kl	days_of_cov
0	SKU_001	Ahmedabad	MHW	A	1.66	2.02	4	1	2.05	8.99	15.64	9
1	SKU_001	Bangalore	MHW	A	1.51	1.7	6	2	2.05	10.56	19.63	12.
2	SKU_001	Bhiwandi	MHW	A	3.61	7.33	3	1	2.05	27.1	37.92	10.
3	SKU_001	Guwahati	MHE	A	0.42	0.57	6	2	2.05	3.34	5.88	13
4	SKU_001	Haryana	MHW	A	8.05	4.37	6	2	2.05	39.71	88.03	10.
5	SKU_001	Hyderabad	MHW	A	5.6	4.51	5	1	2.05	23.7	51.69	9.
6	SKU_001	Jamshedpur	MHE	A	1.31	1.98	4	1	2.05	8.58	13.83	10.
7	SKU_001	Kanpur	MHE	A	11.72	6.96	6	2	2.05	59.5	129.8	11.
8	SKU_001	Kolkata	MHE	A	4.2	2.58	3	1	2.05	12.6	25.21	

Download CFA norms (CSV)

Every SKU × CFA norm — safety stock, reorder point and days of cover from demand + lead-time variability — computed on the fly for all 957 combinations, with one-click CSV download. Hub-level norms (98% service level) sit just below in the app.

08 Robustness & Key Assumptions

Stress-tested against every input-shock type the brief names



Scenario	Status	Total cost	Unmet
Base plan	Optimal	₹9.75 cr	0 kl
BOM capacity -50%	Optimal	₹10.00 cr	0 kl
Demand +40%	Optimal	₹15.95 cr	6.6 kl
BOM→MHW freight x5	Optimal	₹10.86 cr	0 kl

Under a genuine capacity squeeze (demand +40%) the tool leaves only 6.6 kl unmet — and protects contractual SKUs first. It never returns an error.

Assumptions we make — and stand behind

Assumption	Rationale
SKU tiers derived from cumulative volume share (A 50 / B 30 / C 15 / D 5%)	Exhibit F defines tiers by volume; the brief gives penalty & contractual flags, not tiers, so we derive them from 6-month sales.
Safety stock sized from forecast error , not raw sales variance	Next month is planned on a forecast; buffer must protect against forecast error over the lead time.
Contractual SKUs get a 5x penalty multiplier (soft, not hard)	Brief says their breach cost "significantly exceeds normal lost margin"; soft keeps the model solvable even if capacity is critically short. Tunable parameter.
Batch-rounding surplus held as plant closing inventory (flow ≤ production)	25 kl batches force minor over-production; keeping it at the plant is cheaper than freighting it downstream — verified: forcing equality was strictly worse.
One grease SKU (kg-based) normalised to a volume equivalent	99 of 100 SKUs are litre-based; normalising keeps the whole model in one consistent unit (kl).

Bottom line: a rigorous analytical core, a benchmarked least-cost plan that serves 100% of customers, and a robust tool any planner can drive — with every choice documented and defensible.